TEACHING WALLAH

TET Exam

Subject - MATHS

Topic – Algebra

LECTURE NO -01

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Topics to be Covered

Topic Algebra

$$(a+b)^2 = a^2+b^2+2ab$$

$$(a-b)^2 = a^2+b^2-2ab$$

$$a^2-b^2 = (a+b)(a-b)$$

$$(a+b)^3 = a^3+b^3+3ab(a+b)$$

$$(a-b)^3 = a^3-b^3-3ab(a-b)$$

$$a^3+b^3 = (a+b)(a^2+b^2-ab)$$

$$a^3-b^3 = (a-b)(a^2+b^2+ab)$$

$$(a+b+c)^2 = a^2+b^2+c^2+2ab+2bc+2ca$$

If $x+\frac{1}{x}=a$ find $x^2+\frac{1}{x^2}$?

$$\left(x+\frac{1}{x}\right)^2=a^2$$

$$x^2+\frac{1}{x^2}+2\times \cancel{x}\times\frac{1}{\cancel{x}}=a^2$$

$$\boxed{x^2+\frac{1}{x^2}=a^2-2}$$

- $x+\frac{1}{x}=3$ find $x^2+\frac{1}{x^2}$?

$$3^2-2=9-2=7$$

- If $a+\frac{1}{a}=5$ then find $a^2+\frac{1}{a^2}$.

$$5^2-2=25-2$$
$$=23.$$

$x+\frac{1}{x}=a$ find $x^3+\frac{1}{x^3}$?

$$\left(x+\frac{1}{x}\right)^3=a^3$$

$$x^3+\frac{1}{x^3}+3\times \cancel{x}\times\frac{1}{\cancel{x}}\left(x+\frac{1}{x}\right)=a^3$$

$$x^3+\frac{1}{x^3}+3a=a^3$$

$$\boxed{x^3+\frac{1}{x^3}=a^3-3a}$$

If $p+\frac{1}{p}=5$ find

$p^3+\frac{1}{p^3}$?

$=5^3-3(5)$

$=125-15=110$ Ans

If $p+\frac{1}{p}=3$ then find $p^2+\frac{1}{p^2}$ & $p^3+\frac{1}{p^3}$?

- $3^2-2=7$
- $3^3-3(3)=27-9=18$ Ans

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$x+\frac{1}{x}=a$ then find $x^4+\frac{1}{x^4}$?

$$\downarrow$$

• $(a^2-2)^2-2$.

If $x+\frac{1}{x}=3$ find $x^4+\frac{1}{x^4}$?

• $3^2-2=7$

• $7^2-2=47$

$x+\frac{1}{x}=5$ find $x^4+\frac{1}{x^4}$?

$5^2-2=23$

$23^2-2=529-2$

$=527$ Ans

\# If $x+\frac{1}{x}=4$ find $x^9+\frac{1}{x^9}$?

$(x^3)^3 = x^9.$

• $4^3-3(4) = 64-12 = 52$

• $52^3-3(52) = 52(52^2-3]$

$= 52\times(2701)$

$= 140452.$

If $x+\frac{1}{x}=3$ find $x^9+\frac{1}{x^9}$.

$= 3^3-3(3)$

$= 27-9=18$

• $18^3-3(18)$

$= 18(324-3)$

$= 18\times321$

$= 5778$

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If $x+\frac{1}{x}=-2$.

then find. $x^2+\frac{1}{x^2}$; $(-2)^2-2=4-2=2$

- $x^3+\frac{1}{x^3}$; $(-2)^3-3(-2)=-8+6=-2$

- $x^4+\frac{1}{x^4}$; $2^2-2=2$ Ans

- $x^9+\frac{1}{x^9}$; $(-2)^3-3(-2)=-2$ Ans

If $x+\frac{1}{x}=2$; find $x^2+\frac{1}{x^2}$, $x^3+\frac{1}{x^3}$, $x^4+\frac{1}{x^4}$, $x^5+\frac{1}{x^5}$?

$\downarrow$ 2 $\quad\downarrow$ 2 $\quad\downarrow$ 2 $\quad\downarrow$ 2 Ans

$\downarrow$

- $2^2-2=2$
- $2^2-2=2$ Ans

$x+\frac{1}{x}=a$ find $x^6+\frac{1}{x^6}$?

$(x^2)^3$ या $(x^3)^2$.

$(a^2-2)^3-3(a^2-2)$

$x+\frac{1}{x}=3$ find $x^6+\frac{1}{x^6}$?

$\downarrow$

- $3^3-3(3)=27-9=18$
- $18^2-2=322$

$x+\frac{1}{x}=4$ find $x^6+\frac{1}{x^6}$?

- $4^3-3(4)=64-12=52$
- $52^2-2=2702$ Ans

2 minutes Summary

Topic Algebra

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THANK YOU